1a) $S_0(t) = 1 - \left(\frac{t}{120}\right)^2 = \frac{120^2 - t^2}{120^2}$, $0 \le t \le 120$

120 represents limiting age, nobody lives past this age

1b) S(0) = 1

$\lim_{x \to 120} S(x)$ = S(120) = 0

S(x) must be **non-increasing**, $\frac{d}{dx}$ S(x) = -2 $\left(\frac{t}{120}\right)$ ≤ 0. True **because** t is nonnegative.

1c) ${}_tp_x = \frac{S(t+x)}{S(x)} = \frac{\frac{120^2 - (x+t)^2}{120^2}}{\frac{120^2 - x^2}{120^2}} = \frac{120^2 - (x^2 + 2xt + t^2)}{120^2 - x^2} = 1 - \frac{2xt + t^2}{120^2 - x^2}$

0≤x+t≤120

X, age, [0,120]

T, future lifetime, [0,120]

X + T can't exceed omega

We've shown that S_0 is a valid survival model. Assuming consistency ensures that S_y is a valid survival model for y≥0

1d) ${}_tp_{80} = 1 - \frac{160t + t^2}{120^2 - 80^2} = \frac{8000 - 160t - t^2}{8000}$

$E(T_{80}) = \int_0^{40} \frac{8000-160t-t^2}{8000}\,dt = \left.\frac{8000t-80t^2-t^3/3}{8000}\right|_0^{40} =$

$\frac{8000*40-80*40^2-40^3/3}{8000}$ = 21.33 years

1e) $E(T_{80}) = \int_0^{40} S_{80}(t)dt = \int_0^3 S(t)dt + \int_3^{40} S(t)dt = \int_0^3 e^{-\mu t}dt$ under CF mortality + ${}_3p_{80}^{CF} \int_0^{37} S_{83}(t)\,dt$

$= \frac{1-e^{-3\mu}}{\mu} + e^{-3\mu}$ $E(T_{83})$ = 18.10 years

1f) $E(T_{80})$ for Sam (skydiving is a dangerous hobby) should be lower than for Ted.

So this is reasonable. [If students made a calculation error and incorrectly swapped Sam and Ted, they could then say that the constant force of mortality needs to be larger, to receive the mark here]

Possible to receive 1 or 2 points for other appropriate comments.

2a) Before 8th premium, 7 bonuses at 3% have been declared. Benefit for next year is $40{,}000(1.03)^7$

Future bonus = fb = 0.0192308

EPVterm = $40{,}000*(1.03^7)*[\frac{(1+fb)^0}{1.06} * \Pr(K=0) + \frac{(1+fb)}{1.06^2} * \Pr(K=1) + \frac{(1+fb)^2}{1.06^3} * \Pr(K=2) + \cdots + \frac{(1+fb)^{12}}{1.06^{13}} * \Pr(K=12)]$

EPVterm *(1+fb)/(1+fb) and moving numerator (1+fb) inside [] yields $40{,}000 * 1.03^7 * A^1_{47:13}$@4%)/(1+fb) =1728.92 using

$A^1_{47:13} = A_{47:13} - \frac{D_{60}}{D_{47}}$, all at 4% = .60665 - .57083 = .03582

EPVpe will receive 13 future bonuses = $40{,}000*1.03^{7}*\frac{l_{60}}{l_{47}}*\frac{(1+fb)^{13}}{1.06^{13}}$ = 28,082.14

28,082.14 + 1728.92 = 29,811 = EPVFB

2b)

EPVFE = 200 * $A_{\overline{47:13}|}$ at 6% = 200 * .47645 = 95.29

EPVFP after expenses=(1-.02) * 2000 $\ddot{a}_{\overline{47:13}|}$ @6%= .98*2000*9.249 = 18,128

V(7) = EPVFB + EPVFE – EPVFP = 11,248

Some solutions are not provided below. See Excel file for those questions.

3a) See Excel

3bi)

3bii)

3biii) ${}_{.75}q_{K+.5}$

3biv) No, we typically use an assumption between integer ages. It would be better to use μ_K and μ_{K+1} as this probability goes from K+.5 to K+1.25

3c) This life table provides l_x for integer ages, enough to define curtate future lifetime random variables. These assumptions allow us to fully specify the continuous survival model. Hence, we can calculate the probabilities above that don't involve integer values.

3di)

3dii) Not appropriate between 0,3 because newborns are subject to a much higher force of mortality than at age 2. See ELT15 (Males).

Appropriate from 8 to 11, very similar μ. See ELT15 (Males).

Other ages are also acceptable, best if student references a table or a graph in their notes.

4)

5) See Excel

6a) $(\mathrm{I}\ddot{a})_{[51]:4}$

6b) 2 years

6c)